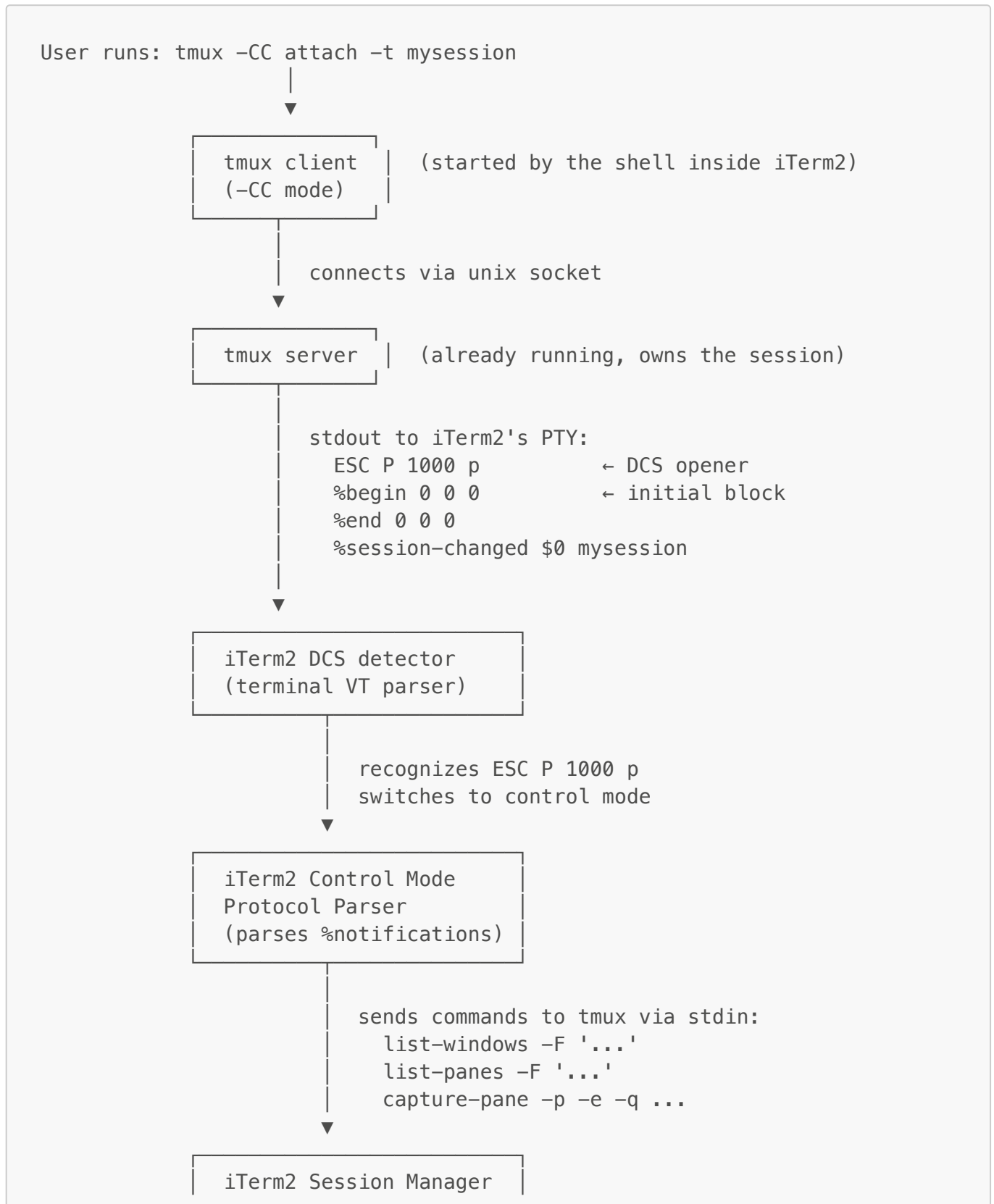


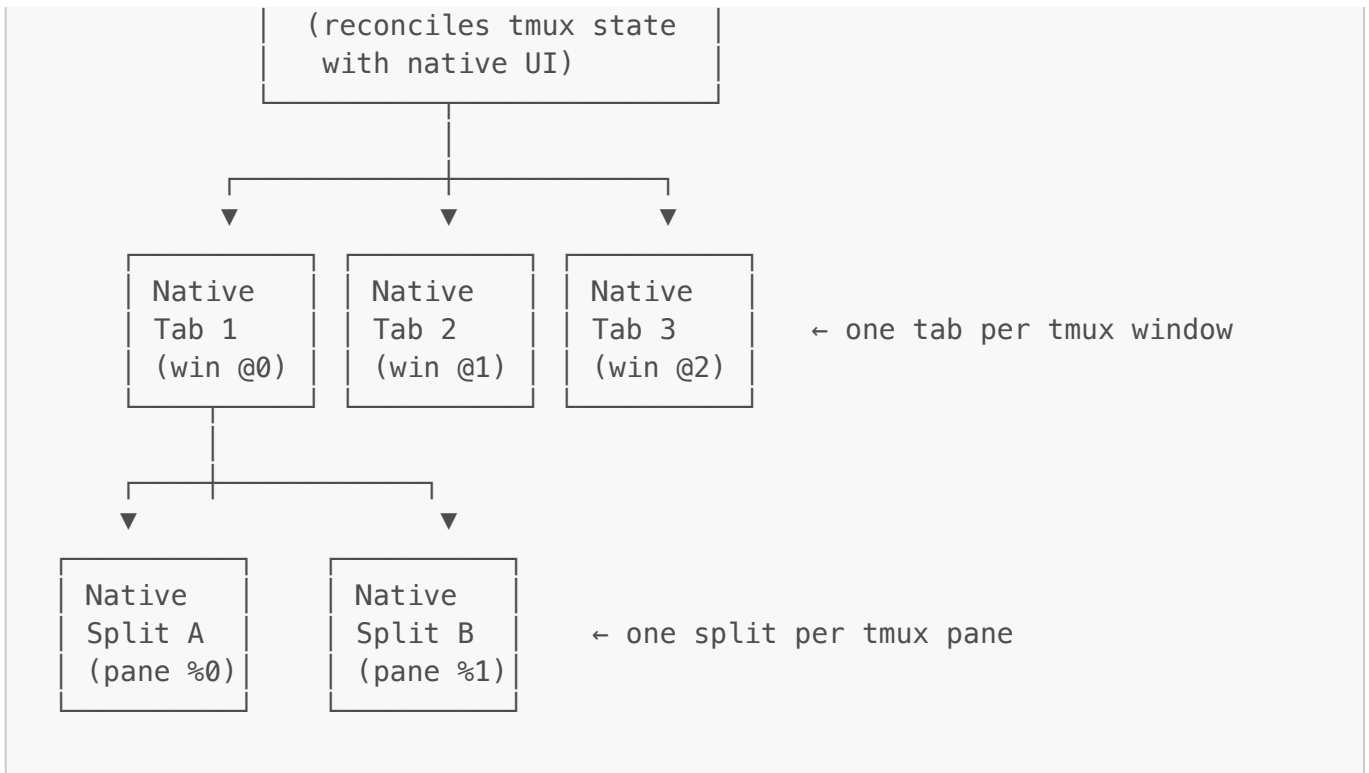
Data Flow Diagrams: tmux Control Mode

1. iTerm2 with tmux Control Mode (Reference Implementation)

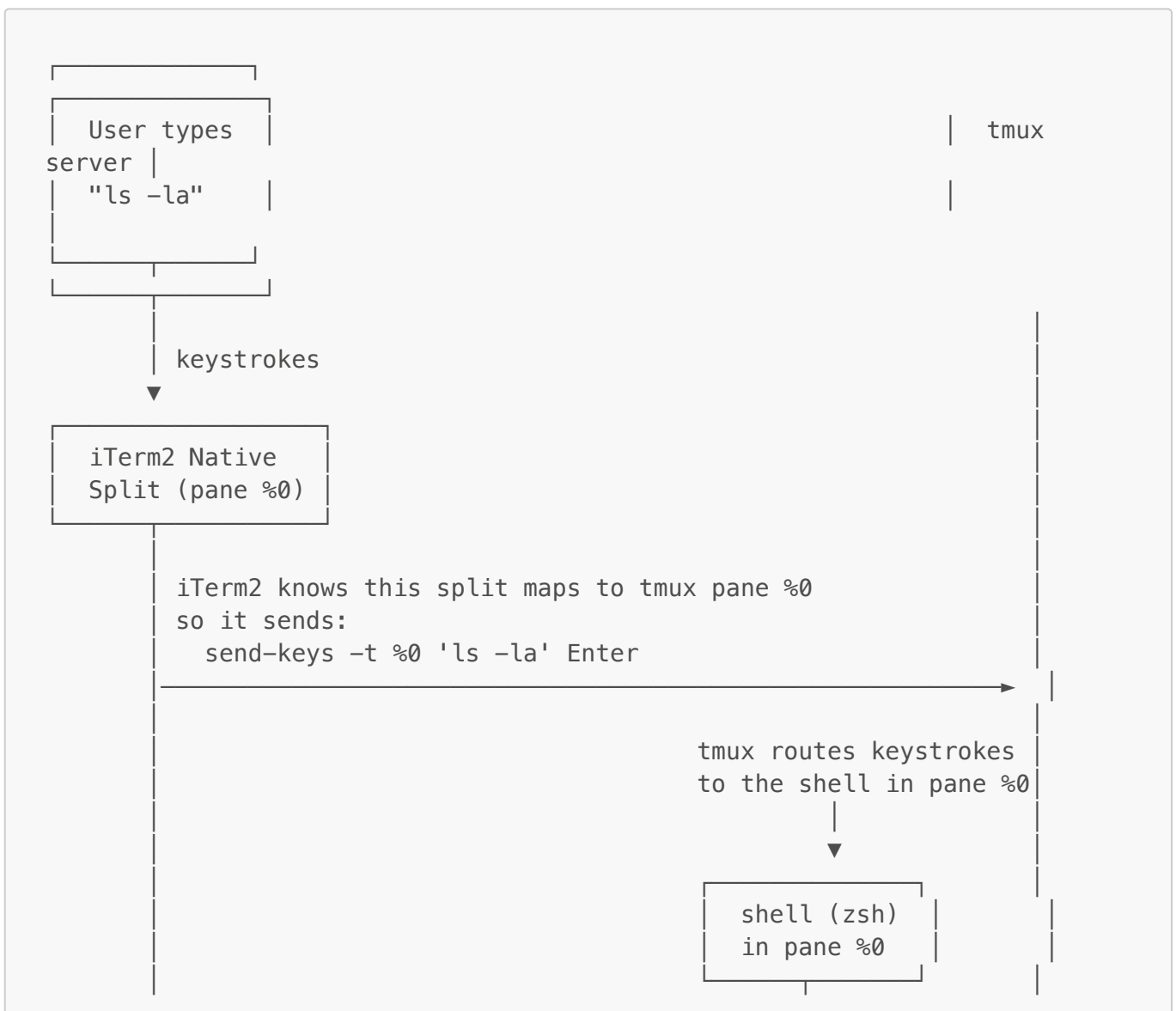
This is how data flows today in a working tmux control mode integration.

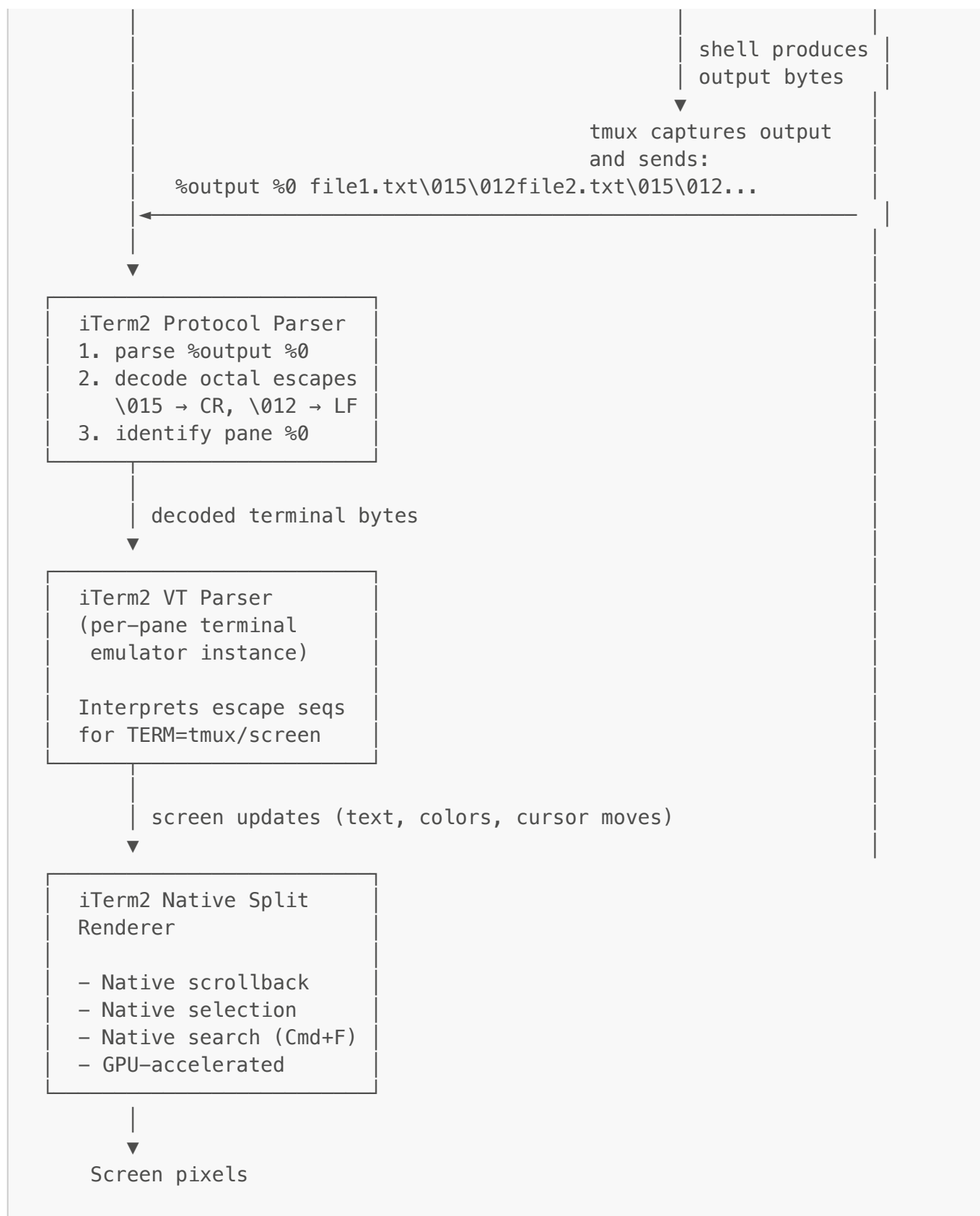
1.1 Startup / Session Discovery



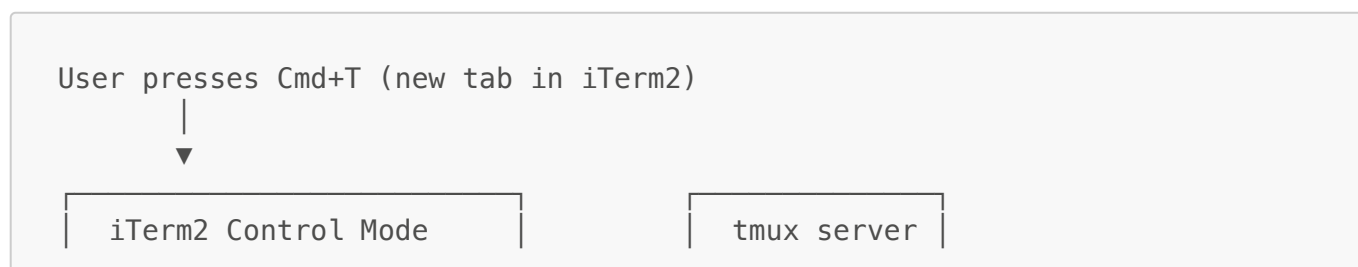


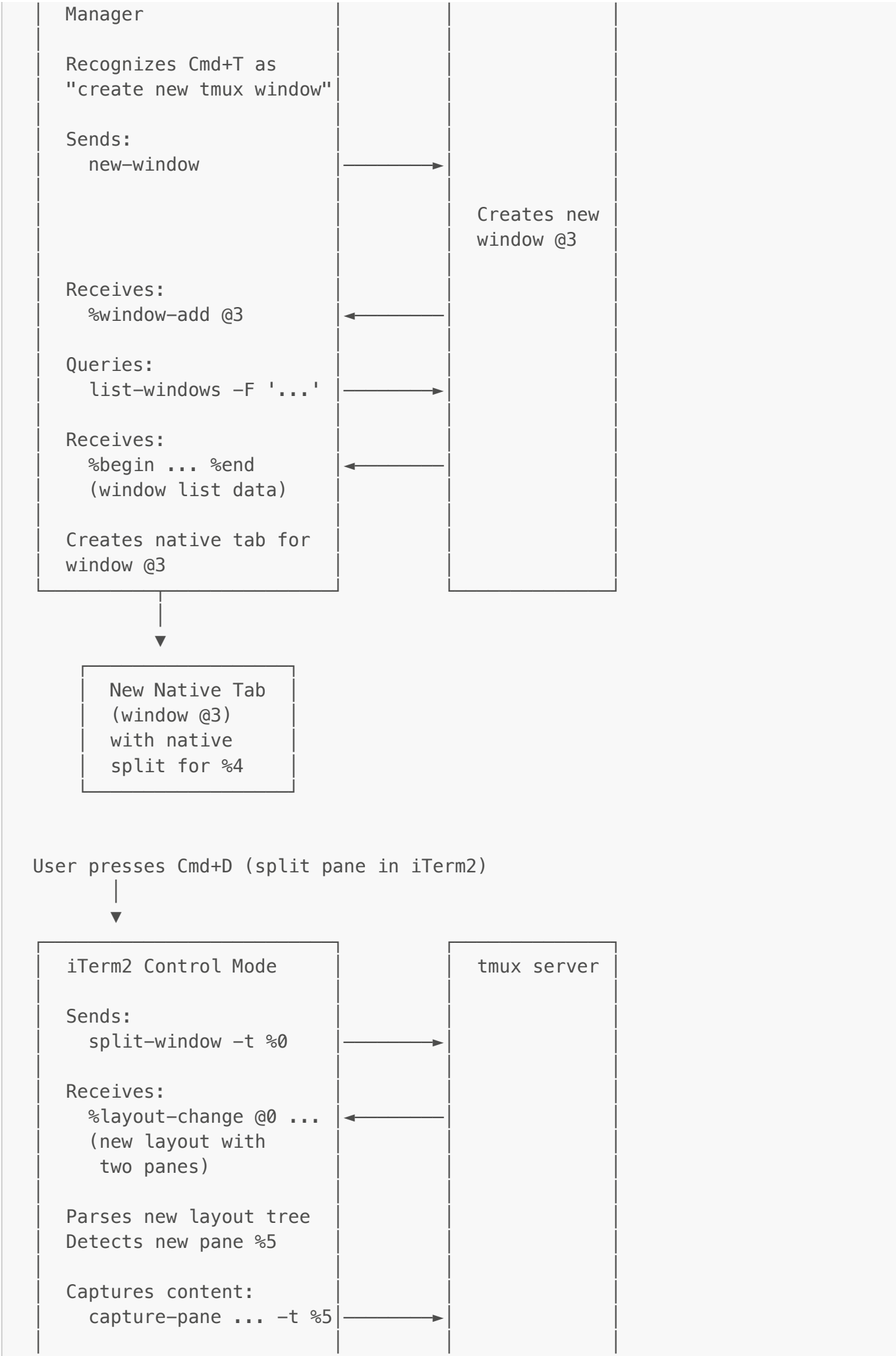
1.2 Steady-State: User Types in a Pane

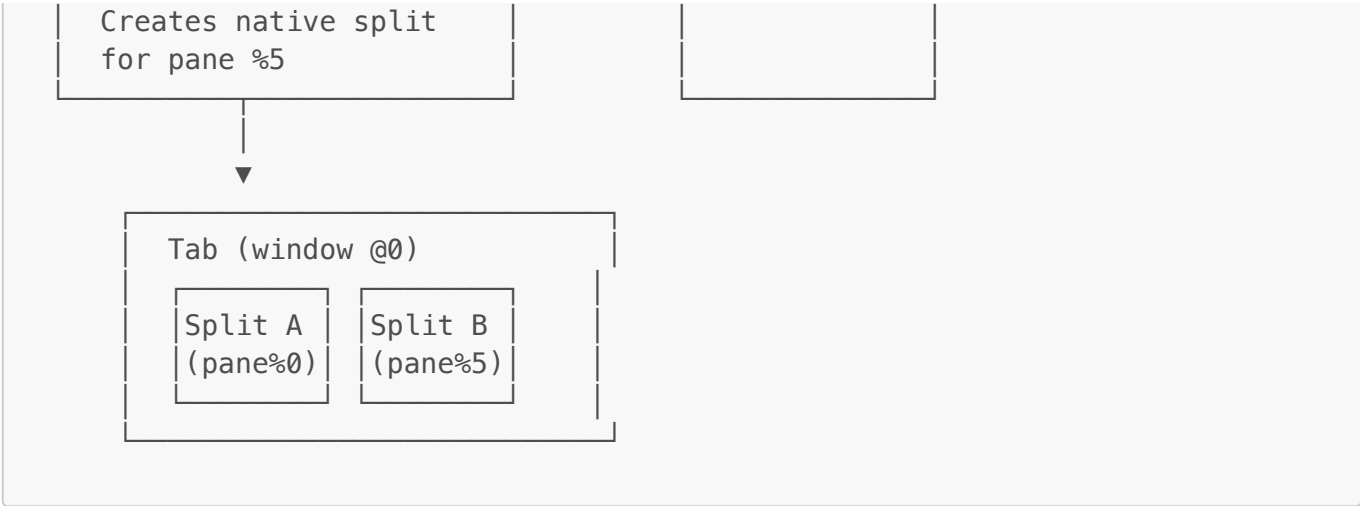




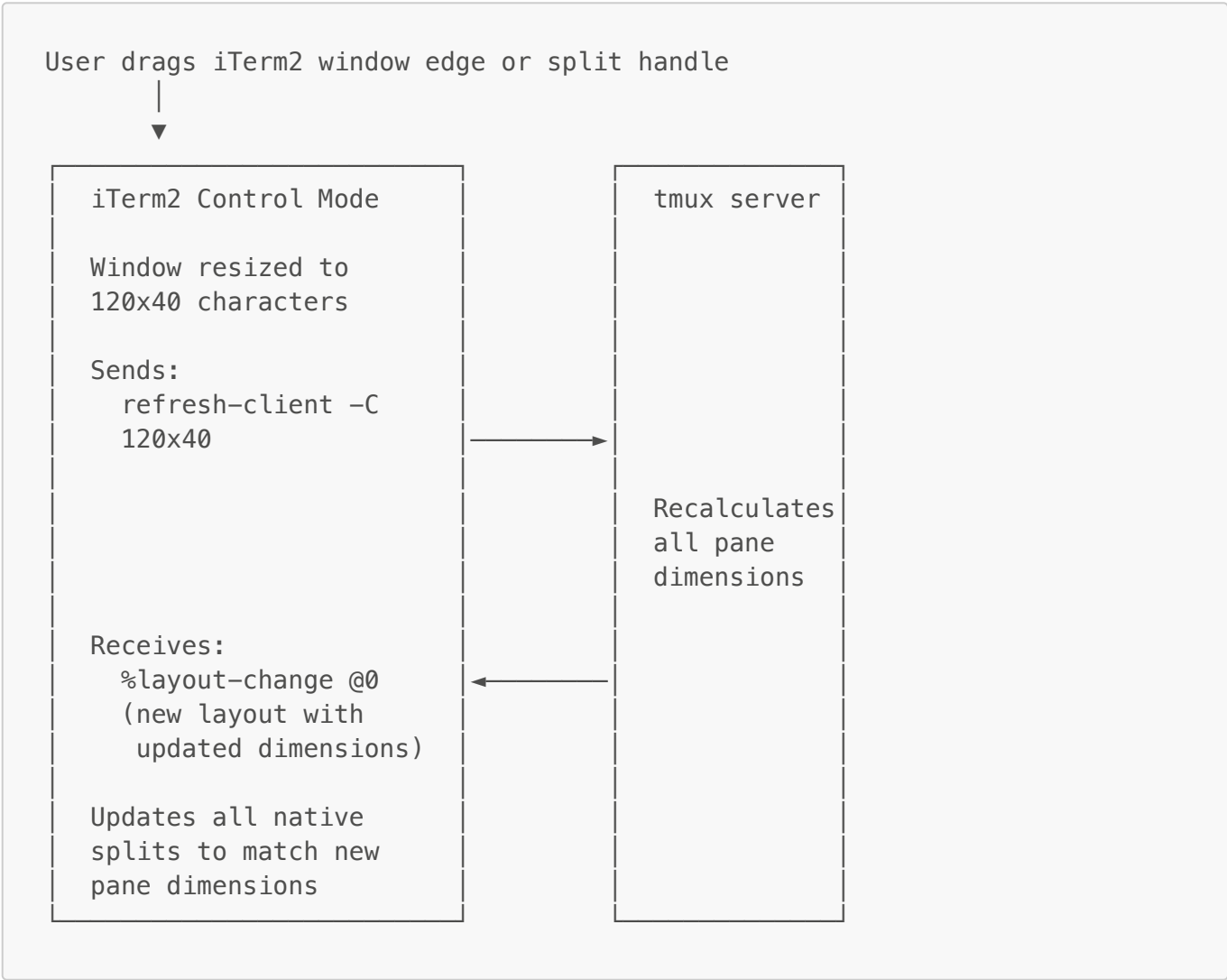
1.3 Steady-State: Structural Changes (New Window, Split, Close)





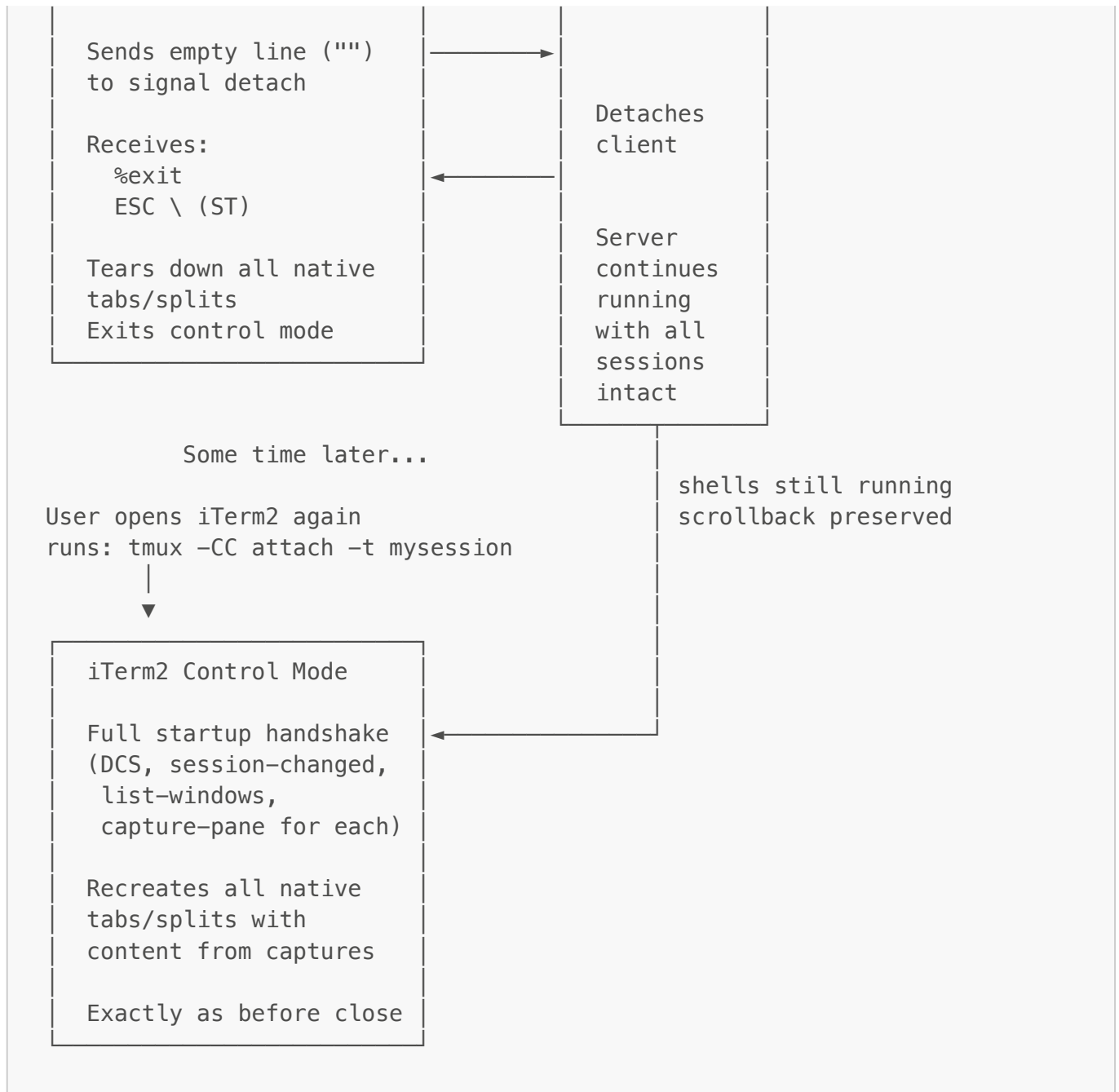


1.4 Resize Flow



1.5 Detach / Reattach Flow

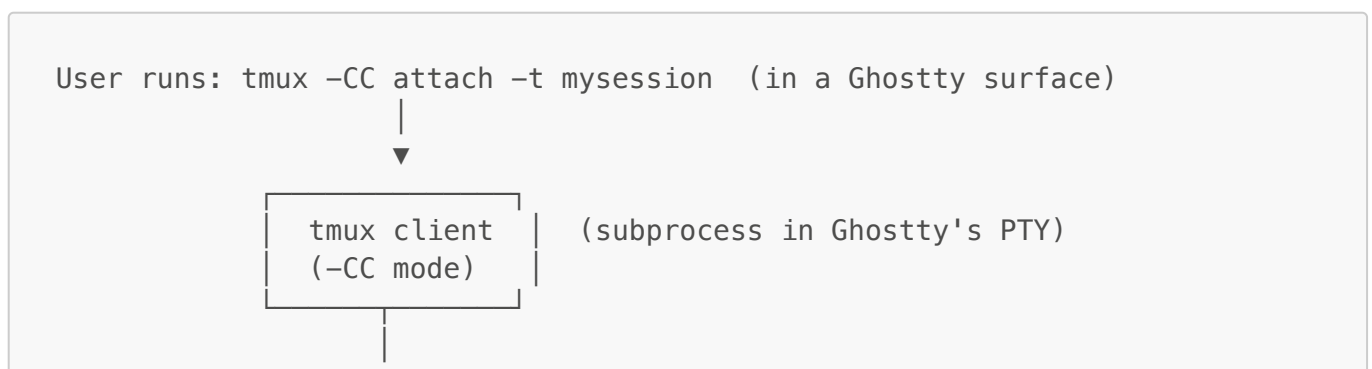


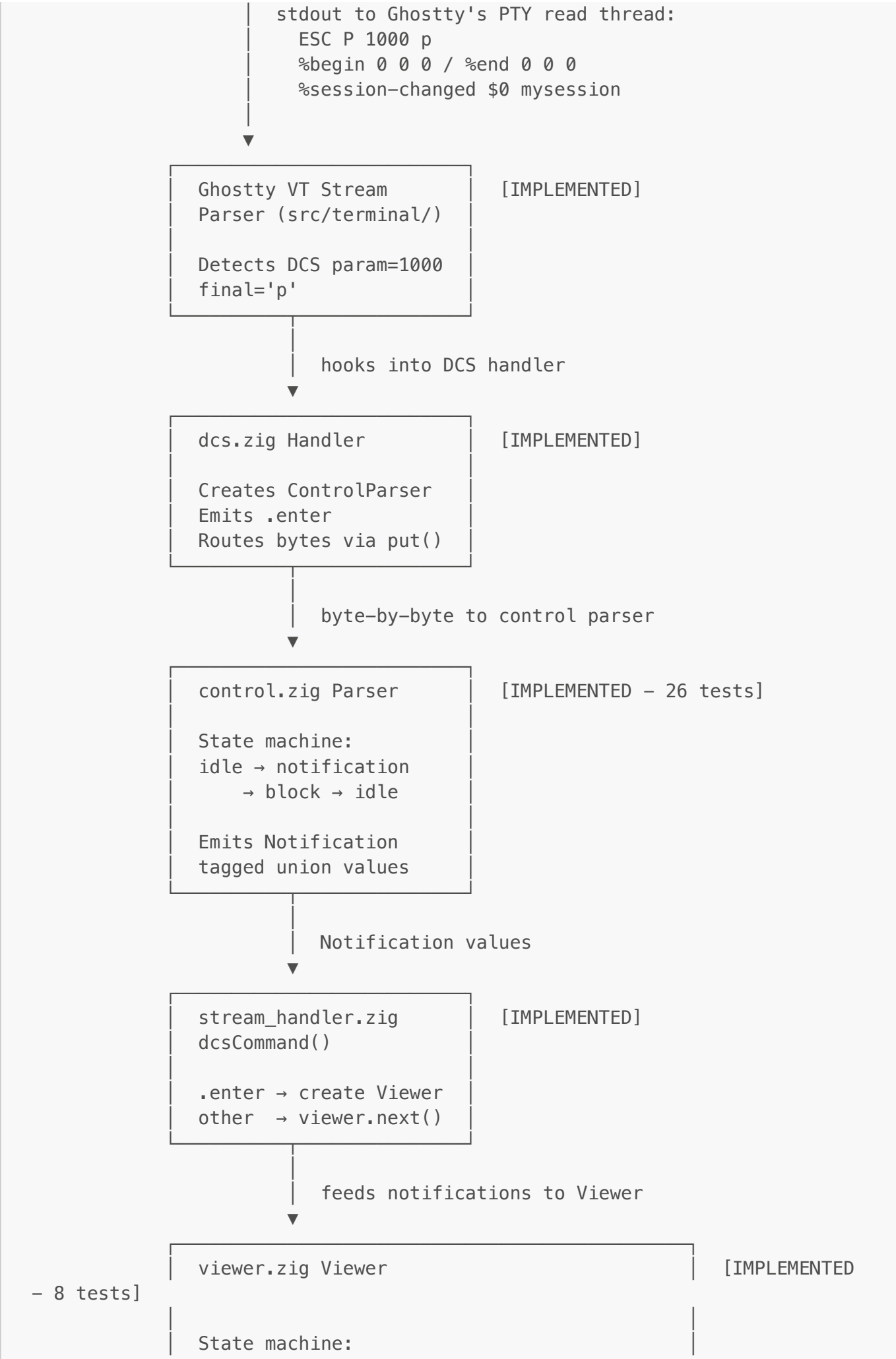


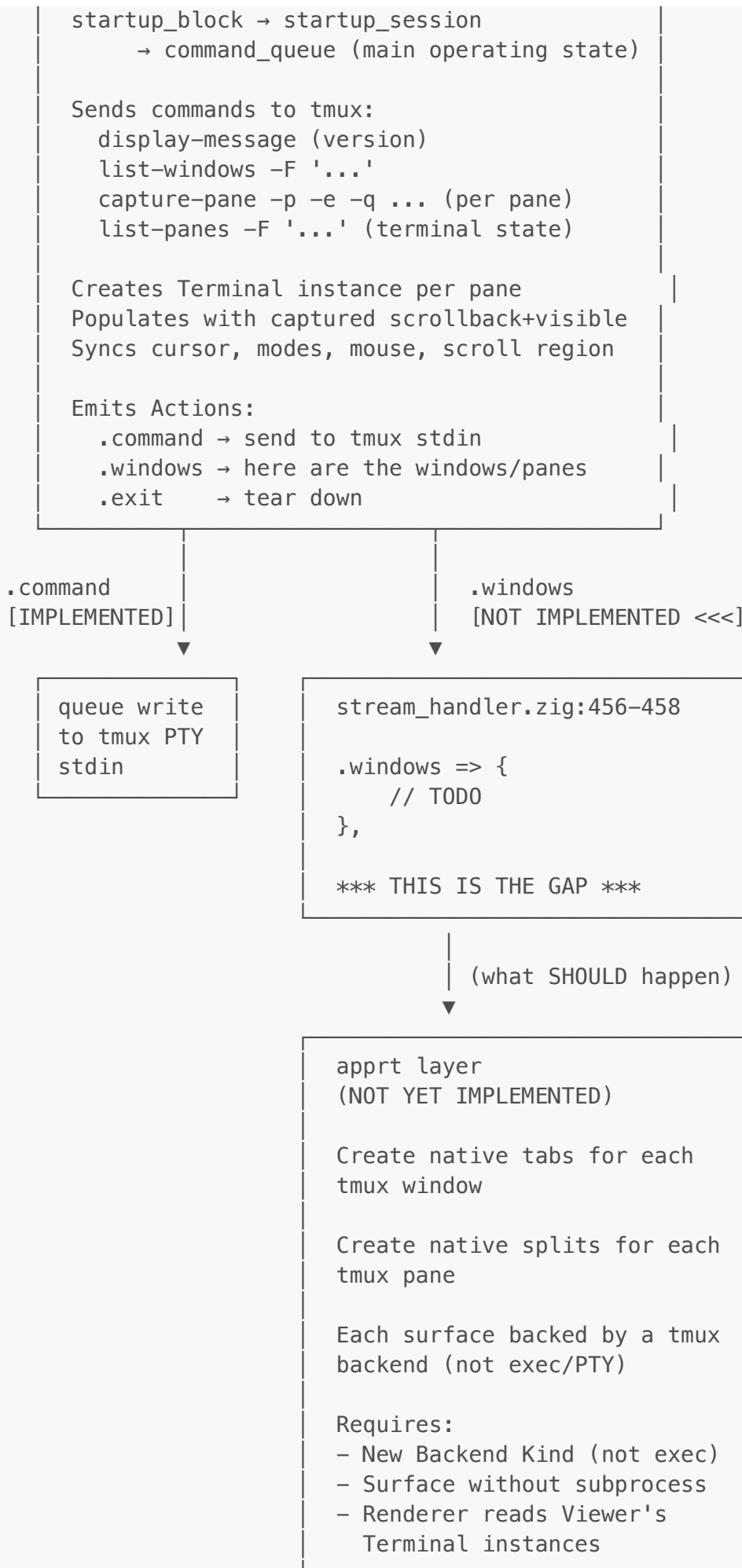
2. Ghostty with tmux Control Mode (Projected Integration)

This shows how data would flow with Ghostty once tmux control mode is fully integrated. Areas that are currently implemented are marked, and the gap is called out.

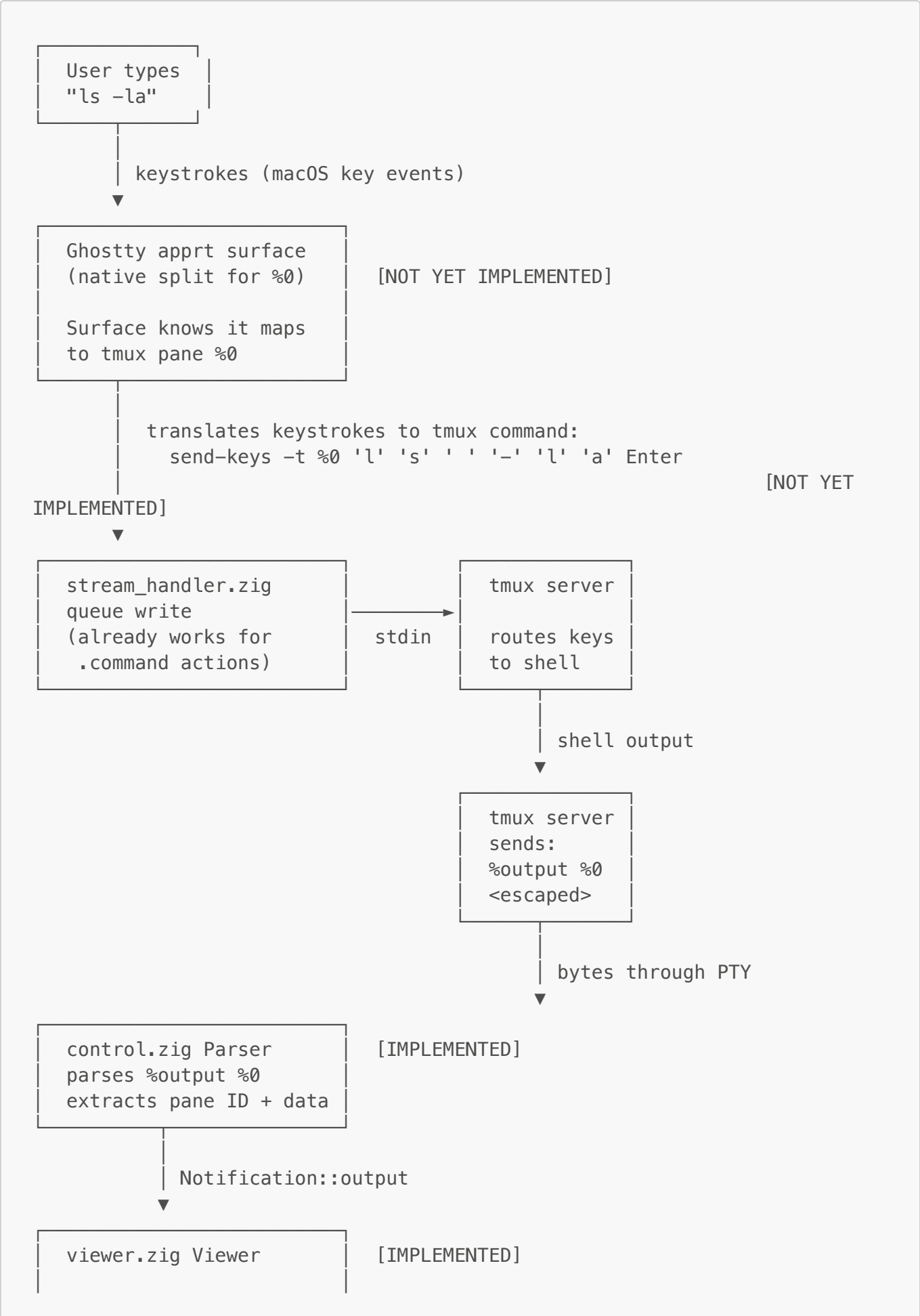
2.1 Startup / Session Discovery

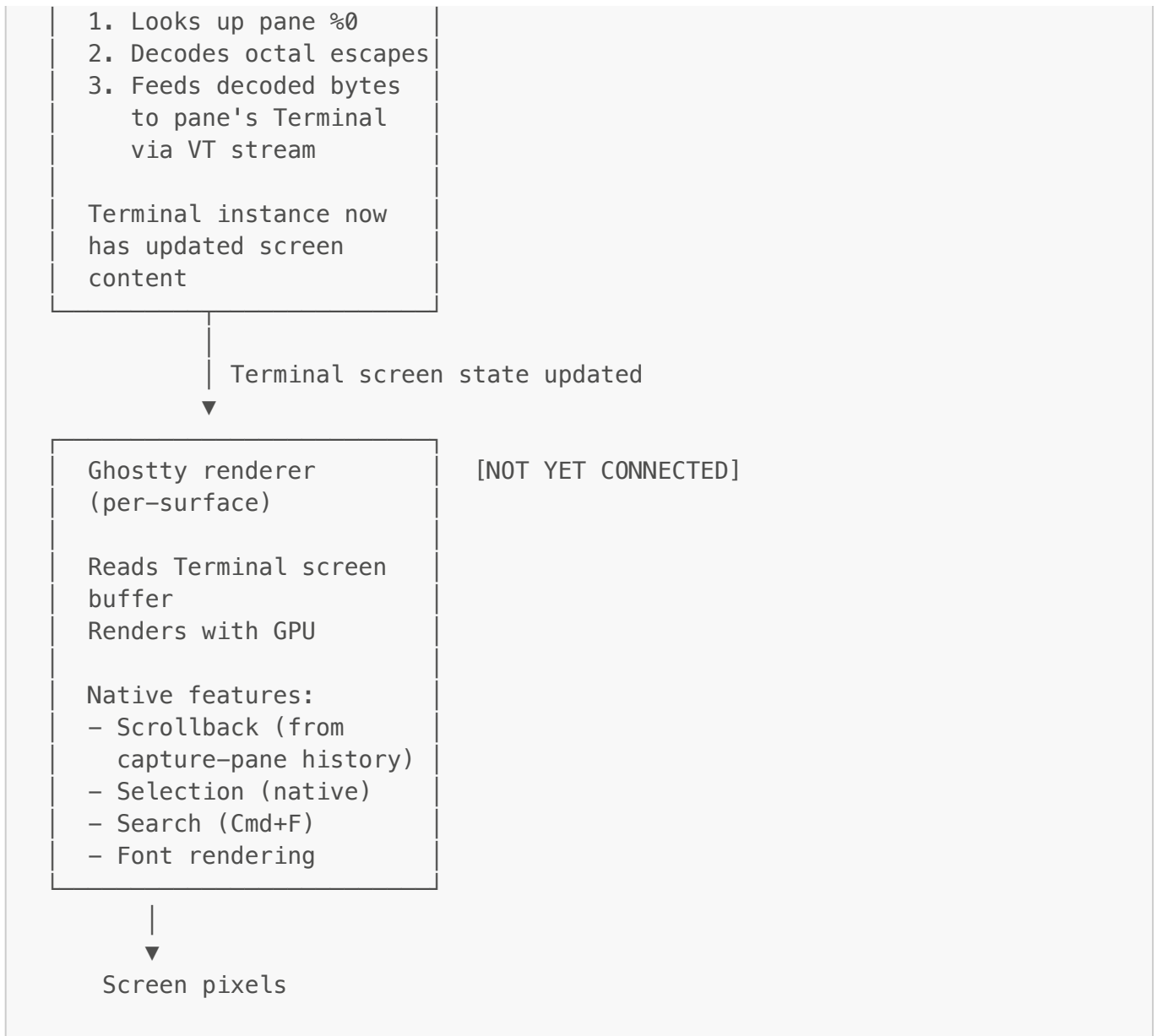




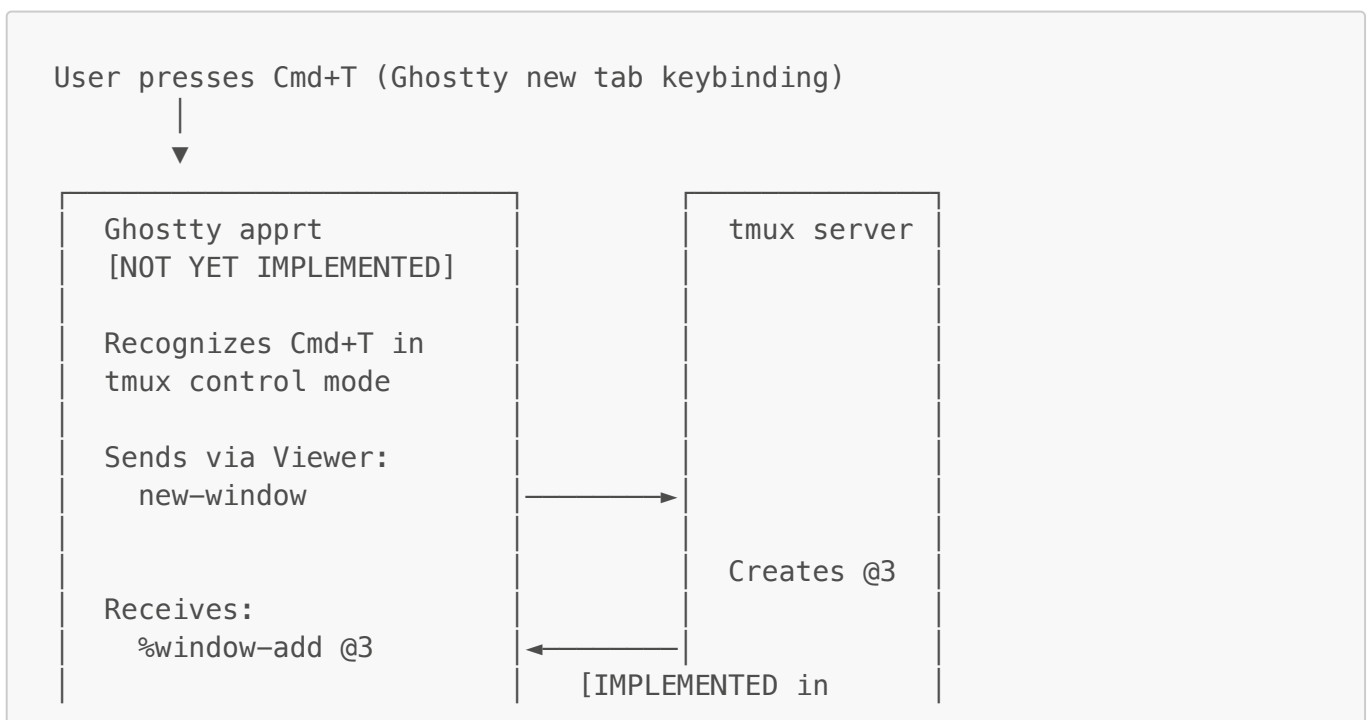


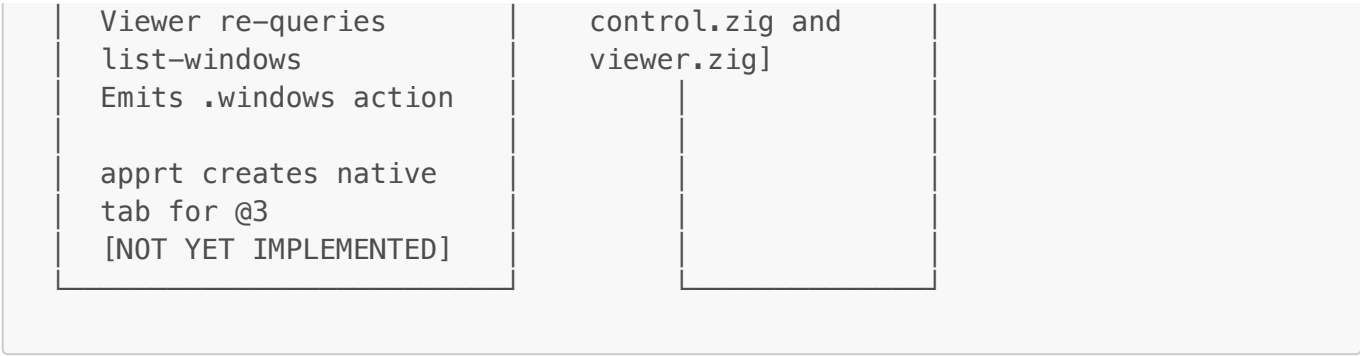
2.2 Steady-State: User Types in a Pane (Projected)



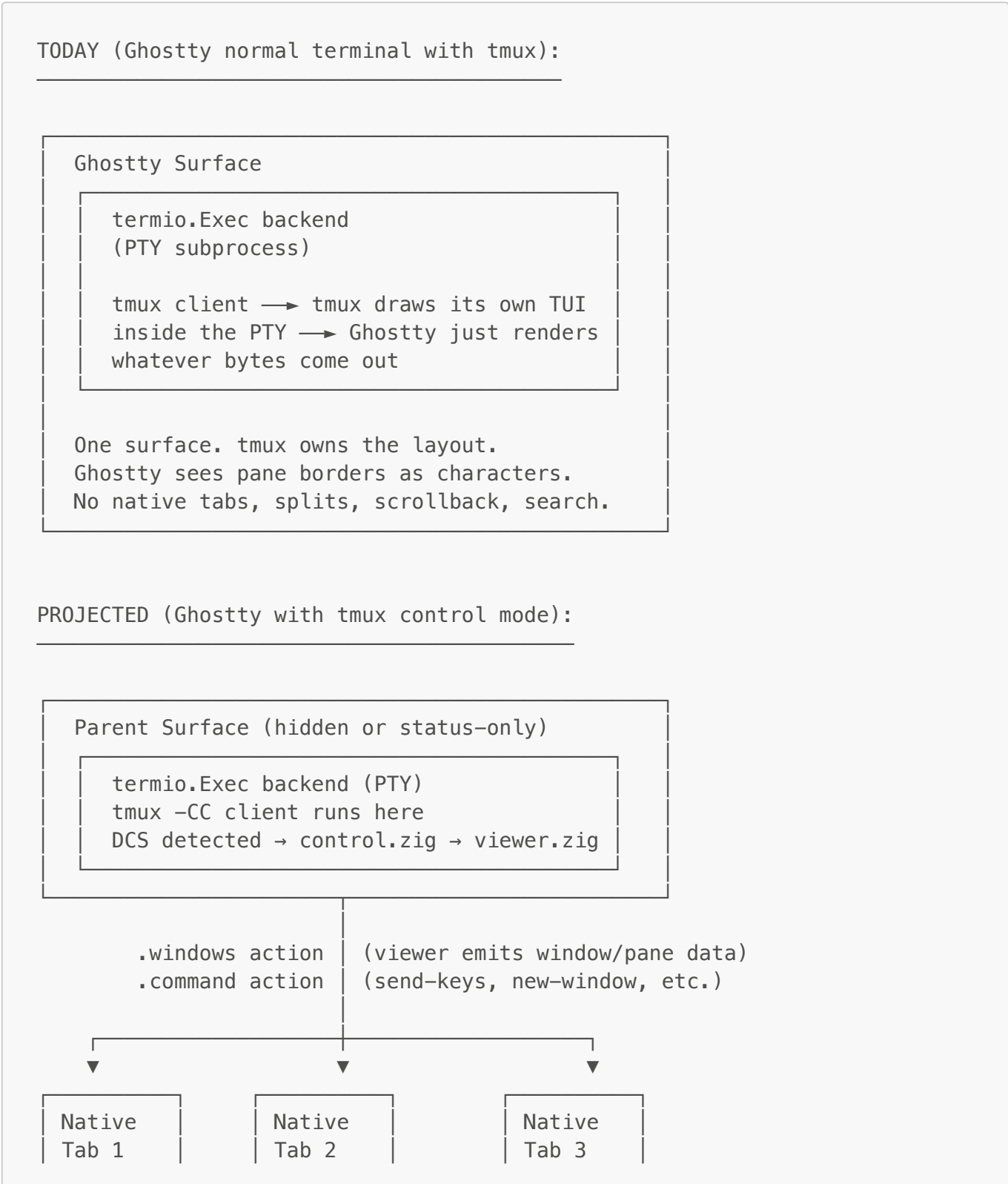


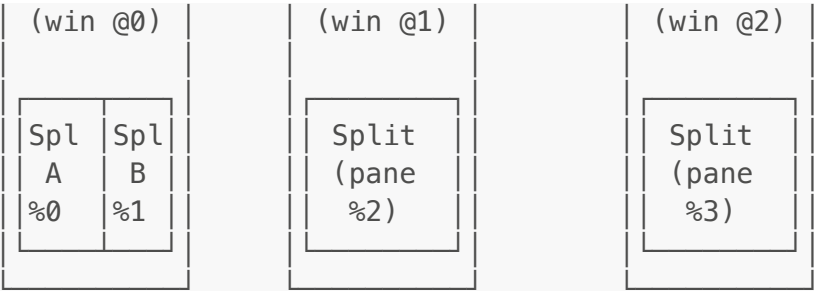
2.3 Steady-State: Structural Changes (Projected)





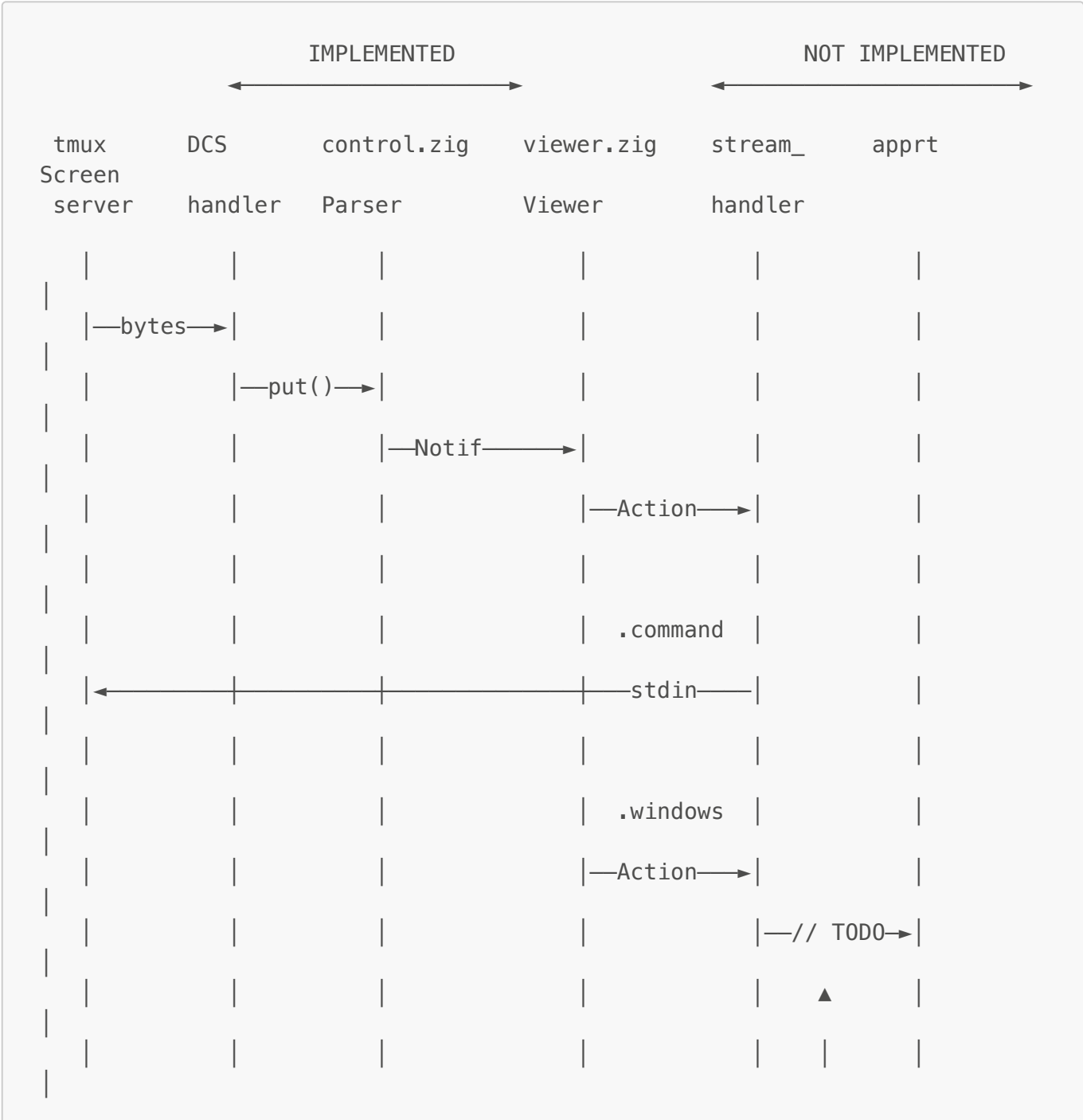
2.4 Architecture Comparison: What Changes





Each split is backed by a tmux Backend (not exec).
Each split's renderer reads the Viewer's Terminal for that pane.
Keystrokes in a split → send-keys -t %<id> → tmux server.
Resize → refresh-client -C WxH → tmux recalculates layout.

2.5 The Integration Gap (Current State)

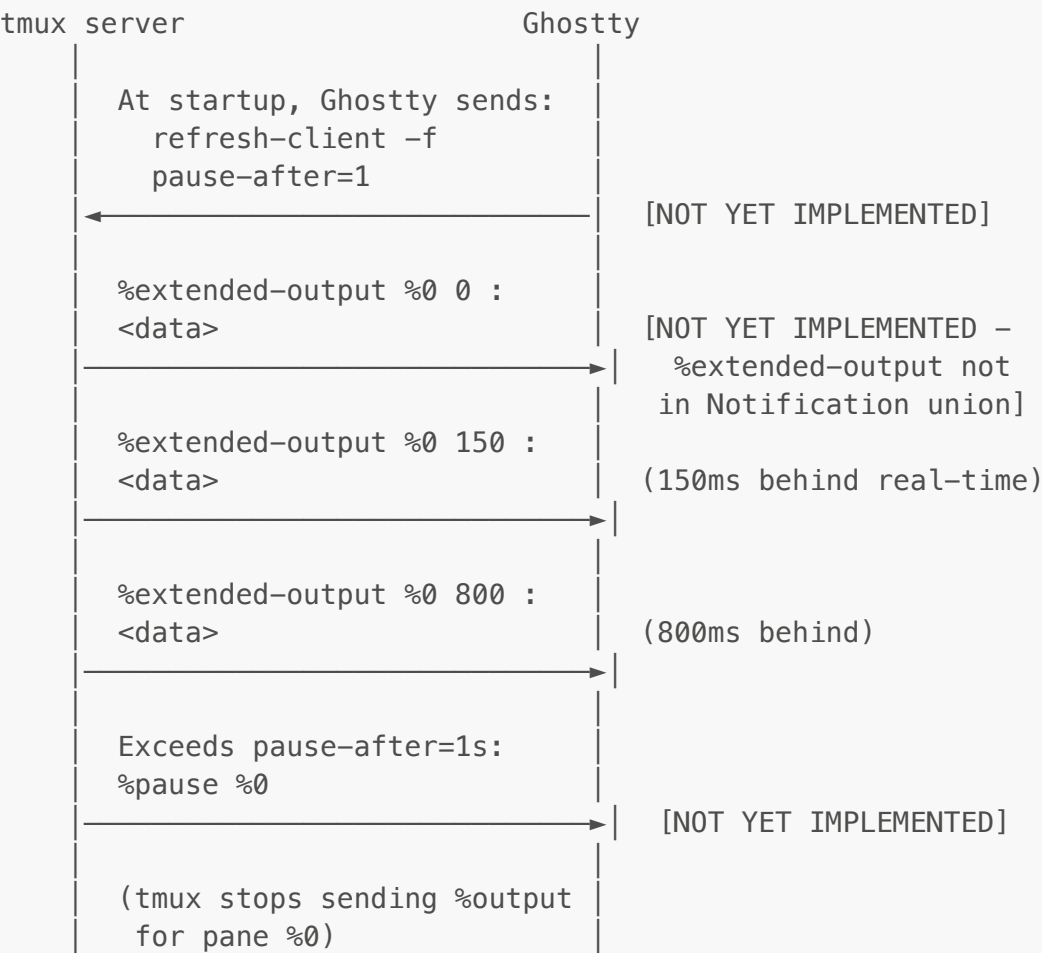


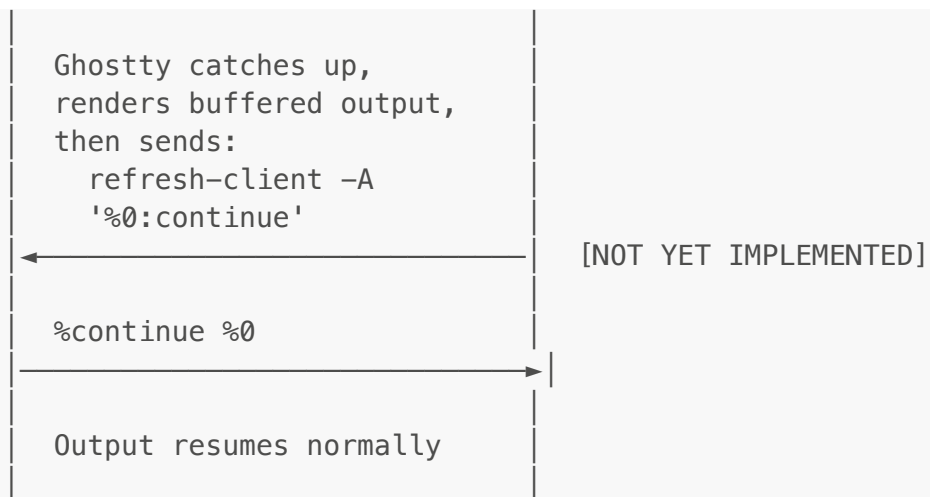
					THE GAP	
					▼	
					Nothing	
					happens	
					here	

- What needs to happen at THE GAP:
- 1. .windows action → create/update native tabs and splits
 - 2. Each surface gets a tmux Backend (new Kind, not exec)
 - 3. Renderer thread reads Viewer's Terminal for that pane
 - 4. Input thread sends send-keys for that pane
 - 5. Resize events send refresh-client -C WxH

2.6 Flow Control (Projected)

Fast-producing pane (e.g., ``find / -name '*.log'``)





Without flow control (current state):

- If client falls >300 seconds behind, tmux disconnects with:
`%exit "too far behind"`
- This is a real risk for fast-producing commands